

DATABASE MANAGEMENT SYSTEM

# **Database Management System for Municipal Waste Collection and Recycling Management System**

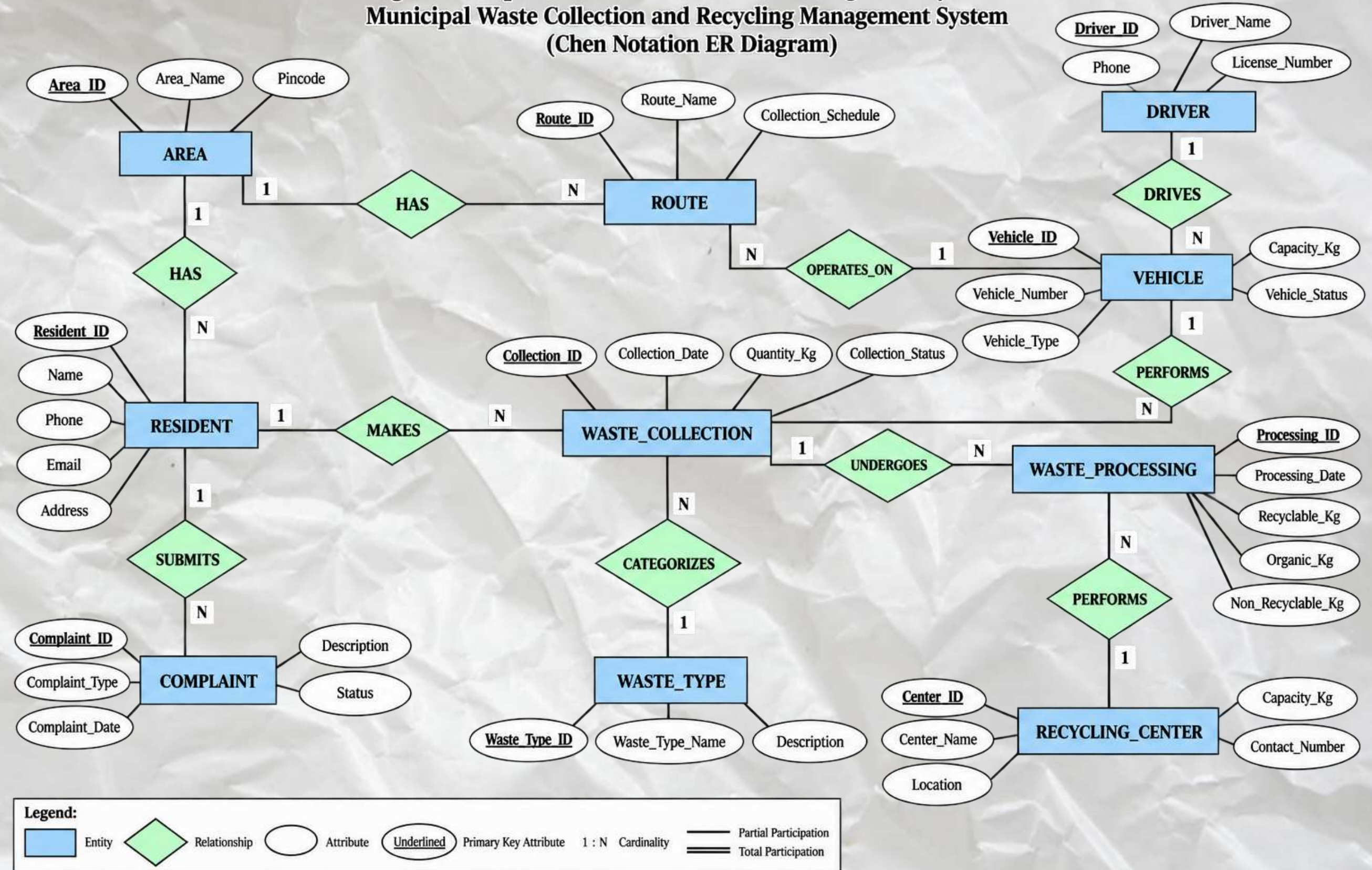
Syed Mohammed Awais Khadiri – 25WU0102283

AIML – WHALES

## Main Entities

- Area
- Resident
- Waste\_Type
- Driver
- Vehicle
- Route
- Waste\_Collection
- Recycling\_Center
- Waste\_Processing
- Complaint

**Design and Implementation of a Database Management System for  
Municipal Waste Collection and Recycling Management System  
(Chen Notation ER Diagram)**



01

12 interconnected  
tables

02

Primary key for  
unique  
identification

03

Foreign key  
for  
relationships

04

NOT NULL, UNIQUE AND  
CHECK constraints

05

DEFAULT values for valid  
values

06

Triggers are automated  
for business rules

## Entities

Area

Resident

Waste\_Type

Driver

Vehicle

Route

Waste\_Collection

Recycling\_Center

Waste\_Processing

Complaint

## Primary Key (PK)

Area\_ID

Resident\_ID

Waste\_Type\_ID

Driver\_ID

Vehicle\_ID

Route\_ID

Collection\_ID

Center\_ID

Processing\_ID

Complaint\_ID

## Foreign Key (FK)

—

Area\_ID

—

—

Driver\_ID

Area\_ID, Vehicle\_ID

Resident\_ID, Vehicle\_ID, Waste\_Type\_ID

—

Collection\_ID, Center\_ID

Resident\_ID

ER Diagram

DB Design

Initial Relational Schema

Filled-in tables in RDBMS

Are

Resident

```
mysql> SELECT * FROM Area;
```

| Area_ID | Area_Name      | Pincode |
|---------|----------------|---------|
| 1       | Miyapur        | 500049  |
| 2       | Kukatpally     | 500072  |
| 3       | Bachupally     | 500090  |
| 4       | Nizampet       | 500090  |
| 5       | Pragathi Nagar | 500090  |
| 6       | Chandanagar    | 500050  |
| 7       | Madinaguda     | 500049  |
| 8       | Hafeezpet      | 500049  |
| 9       | Kondapur       | 500084  |
| 10      | Madhapur       | 500081  |
| 11      | Gachibowli     | 500032  |
| 12      | Manikonda      | 500089  |
| 13      | Narsingi       | 500075  |
| 14      | Khajaguda      | 500008  |
| 15      | Jubilee Hills  | 500033  |
| 16      | Banjara Hills  | 500034  |
| 17      | Ameerpet       | 500016  |
| 18      | SR Nagar       | 500038  |
| 19      | Moosapet       | 500018  |
| 20      | Balanagar      | 500037  |
| 21      | Begumpet       | 500016  |
| 22      | Somajiguda     | 500082  |
| 23      | Punjagutta     | 500082  |
| 24      | Koti           | 500095  |
| 25      | Abids          | 500001  |
| 26      | Himayatnagar   | 500029  |
| 27      | Dilsukhnagar   | 500060  |
| 28      | LB Nagar       | 500074  |
| 29      | Uppal          | 500039  |
| 30      | Tarnaka        | 500017  |

```
mysql> SELECT * FROM Resident;
```

| Resident_ID | Name           | Phone      | Email                    | Address                  | Area_ID |
|-------------|----------------|------------|--------------------------|--------------------------|---------|
| 1           | Aarav Sharma   | 9876543210 | aarav.sharma@gmail.com   | House 12, Miyapur        | 1       |
| 2           | Aisha Khan     | 9876543211 | aisha.khan@gmail.com     | Flat 204, Kukatpally     | 2       |
| 3           | Rahul Reddy    | 9876543212 | rahul.reddy@gmail.com    | House 45, Bachupally     | 3       |
| 4           | Sneha Rao      | 9876543213 | sneha.rao@gmail.com      | Plot 18, Nizampet        | 4       |
| 5           | Vikram Singh   | 9876543214 | vikram.singh@gmail.com   | House 27, Pragathi Nagar | 5       |
| 6           | Ananya Iyer    | 9876543215 | ananya.iyer@gmail.com    | Flat 302, Chandanagar    | 6       |
| 7           | Mohammed Arif  | 9876543216 | mohammed.arif@gmail.com  | House 16, Madinaguda     | 7       |
| 8           | Priya Verma    | 9876543217 | priya.verma@gmail.com    | Flat 105, Hafeezpet      | 8       |
| 9           | Rohan Mehta    | 9876543218 | rohan.mehta@gmail.com    | Villa 22, Kondapur       | 9       |
| 10          | Kavya Nair     | 9876543219 | kavya.nair@gmail.com     | Flat 504, Madhapur       | 10      |
| 11          | Aditya Kapoor  | 9876543220 | aditya.kapoor@gmail.com  | House 33, Gachibowli     | 11      |
| 12          | Fatima Begum   | 9876543221 | fatima.begum@gmail.com   | Villa 11, Manikonda      | 12      |
| 13          | Sahil Gupta    | 9876543222 | sahil.gupta@gmail.com    | Flat 208, Narsingi       | 13      |
| 14          | Neha Joshi     | 9876543223 | neha.joshi@gmail.com     | House 19, Khajaguda      | 14      |
| 15          | Imran Ahmed    | 9876543224 | imran.ahmed@gmail.com    | House 41, Jubilee Hills  | 15      |
| 16          | Meera Krishnan | 9876543225 | meera.krishnan@gmail.com | Villa 8, Banjara Hills   | 16      |
| 17          | Karthik Varma  | 9876543226 | karthik.varma@gmail.com  | Flat 307, Ameerpet       | 17      |
| 18          | Sana Ali       | 9876543227 | sana.ali@gmail.com       | House 25, SR Nagar       | 18      |
| 19          | Ravi Kumar     | 9876543228 | ravi.kumar@gmail.com     | House 52, Moosapet       | 19      |
| 20          | Divya Reddy    | 9876543229 | divya.reddy@gmail.com    | House 14, Balanagar      | 20      |
| 21          | Arjun Malhotra | 9876543230 | arjun.malhotra@gmail.com | Flat 401, Begumpet       | 21      |
| 22          | Pooja Sharma   | 9876543231 | pooja.sharma@gmail.com   | Flat 203, Somajiguda     | 22      |
| 23          | Zaid Hussain   | 9876543232 | zaid.hussain@gmail.com   | House 31, Punjagutta     | 23      |
| 24          | Lakshmi Devi   | 9876543233 | lakshmi.devi@gmail.com   | House 17, Koti           | 24      |
| 25          | Nikhil Bansal  | 9876543234 | nikhil.bansal@gmail.com  | Flat 105, Abids          | 25      |
| 26          | Swathi Rao     | 9876543235 | swathi.rao@gmail.com     | House 29, Himayatnagar   | 26      |
| 27          | Manoj Yadav    | 9876543236 | manoj.yadav@gmail.com    | House 63, Dilsukhnagar   | 27      |
| 28          | Harini Reddy   | 9876543237 | harini.reddy@gmail.com   | Flat 302, LB Nagar       | 28      |
| 29          | Sameer Khan    | 9876543238 | sameer.khan@gmail.com    | House 21, Uppal          | 29      |
| 30          | Ishita Das     | 9876543239 | ishita.das@gmail.com     | Flat 204, Tarnaka        | 30      |

## Waste\_Type

```
mysql> SELECT * FROM Waste_Type;
```

| Waste_Type_ID | Waste_Type_Name      | Description                                                          |
|---------------|----------------------|----------------------------------------------------------------------|
| 1             | Food Waste           | Leftover food, kitchen scraps and other biodegradable food materials |
| 2             | Garden Waste         | Leaves, grass, branches and other garden materials                   |
| 3             | Paper Waste          | Newspapers, notebooks, office paper and cardboard                    |
| 4             | Cardboard            | Packaging boxes and corrugated cardboard materials                   |
| 5             | Plastic Bottles      | PET and other plastic beverage bottles                               |
| 6             | Plastic Packaging    | Plastic bags, wrappers and packaging materials                       |
| 7             | Glass                | Glass bottles, jars and other glass containers                       |
| 8             | Metal Cans           | Aluminium and steel cans used for food and beverages                 |
| 9             | Scrap Metal          | Discarded metal objects and metallic household items                 |
| 10            | Textile Waste        | Old clothes, fabrics and other textile materials                     |
| 11            | E-Waste              | Discarded electronic and electrical equipment                        |
| 12            | Batteries            | Used household and rechargeable batteries                            |
| 13            | Light Bulbs          | Discarded LED, CFL and other light bulbs                             |
| 14            | Used Electronics     | Old computers, phones, televisions and electronic devices            |
| 15            | Hazardous Waste      | Waste containing harmful or toxic substances                         |
| 16            | Paint Waste          | Unused or discarded paints and paint containers                      |
| 17            | Chemical Waste       | Discarded household or industrial chemicals                          |
| 18            | Medical Waste        | Waste generated from medical and healthcare activities               |
| 19            | Sanitary Waste       | Diapers, sanitary pads and other personal hygiene waste              |
| 20            | Construction Waste   | Concrete, bricks, tiles and construction debris                      |
| 21            | Wood Waste           | Discarded wooden furniture, boards and wood materials                |
| 22            | Rubber Waste         | Discarded rubber products and materials                              |
| 23            | Tyres                | Used and discarded vehicle tyres                                     |
| 24            | Used Oil             | Waste cooking oil and used automotive oil                            |
| 25            | Organic Waste        | Biodegradable waste from plants and animals                          |
| 26            | Mixed Waste          | Waste containing multiple waste categories                           |
| 27            | Ceramic Waste        | Broken ceramic plates, cups and tiles                                |
| 28            | Leather Waste        | Discarded leather products and scraps                                |
| 29            | E-Waste Accessories  | Chargers, cables, adapters and electronic accessories                |
| 30            | Non-Recyclable Waste | Waste that cannot be economically recycled                           |

## Driver

```
mysql> SELECT * FROM Driver;
```

| Driver_ID | Driver_Name     | Phone      | License_Number |
|-----------|-----------------|------------|----------------|
| 1         | Ramesh Kumar    | 9000000001 | TS01DL1001     |
| 2         | Suresh Reddy    | 9000000002 | TS01DL1002     |
| 3         | Mahesh Yadav    | 9000000003 | TS01DL1003     |
| 4         | Naveen Kumar    | 9000000004 | TS01DL1004     |
| 5         | Ravi Teja       | 9000000005 | TS01DL1005     |
| 6         | Imran Khan      | 9000000006 | TS01DL1006     |
| 7         | Arun Kumar      | 9000000007 | TS01DL1007     |
| 8         | Prakash Rao     | 9000000008 | TS01DL1008     |
| 9         | Venkatesh Naidu | 9000000009 | TS01DL1009     |
| 10        | Sanjay Singh    | 9000000010 | TS01DL1010     |
| 11        | Rajesh Kumar    | 9000000011 | TS01DL1011     |
| 12        | Anil Reddy      | 9000000012 | TS01DL1012     |
| 13        | Mohan Das       | 9000000013 | TS01DL1013     |
| 14        | Kiran Kumar     | 9000000014 | TS01DL1014     |
| 15        | Salman Ahmed    | 9000000015 | TS01DL1015     |
| 16        | Dinesh Rao      | 9000000016 | TS01DL1016     |
| 17        | Ajay Sharma     | 9000000017 | TS01DL1017     |
| 18        | Ganesh Yadav    | 9000000018 | TS01DL1018     |
| 19        | Harish Kumar    | 9000000019 | TS01DL1019     |
| 20        | Raghu Varma     | 9000000020 | TS01DL1020     |
| 21        | Surya Prakash   | 9000000021 | TS01DL1021     |
| 22        | Naresh Babu     | 9000000022 | TS01DL1022     |
| 23        | Vijay Kumar     | 9000000023 | TS01DL1023     |
| 24        | Abdul Rahman    | 9000000024 | TS01DL1024     |
| 25        | Manoj Kumar     | 9000000025 | TS01DL1025     |
| 26        | Lokesh Reddy    | 9000000026 | TS01DL1026     |
| 27        | Srikanth Rao    | 9000000027 | TS01DL1027     |
| 28        | Chandra Sekhar  | 9000000028 | TS01DL1028     |
| 29        | Bharath Kumar   | 9000000029 | TS01DL1029     |
| 30        | Madan Mohan     | 9000000030 | TS01DL1030     |

ER Diagram

DB Design

Initial Relational Schema

Filled-in tables in RDBMS

# Vehicle

```
mysql> SELECT * FROM Vehicle;
```

| Vehicle_ID | Vehicle_Number | Vehicle_Type    | Capacity_Kg | Driver_ID | Vehicle_Status |
|------------|----------------|-----------------|-------------|-----------|----------------|
| 1          | TS09EA1001     | Mini Tipper     | 1000.00     | 1         | Active         |
| 2          | TS09EA1002     | Garbage Truck   | 5000.00     | 2         | Active         |
| 3          | TS09EA1003     | Compactor Truck | 8000.00     | 3         | Active         |
| 4          | TS09EA1004     | Mini Tipper     | 1500.00     | 4         | Active         |
| 5          | TS09EA1005     | Garbage Truck   | 6000.00     | 5         | Maintenance    |
| 6          | TS09EA1006     | Recycling Truck | 4000.00     | 6         | Active         |
| 7          | TS09EA1007     | Mini Tipper     | 1200.00     | 7         | Active         |
| 8          | TS09EA1008     | Compactor Truck | 9000.00     | 8         | Active         |
| 9          | TS09EA1009     | Garbage Truck   | 7000.00     | 9         | Active         |
| 10         | TS09EA1010     | Recycling Truck | 4500.00     | 10        | Active         |
| 11         | TS09EA1011     | Mini Tipper     | 1000.00     | 11        | Maintenance    |
| 12         | TS09EA1012     | Garbage Truck   | 5500.00     | 12        | Active         |
| 13         | TS09EA1013     | Compactor Truck | 8500.00     | 13        | Active         |
| 14         | TS09EA1014     | Mini Tipper     | 1300.00     | 14        | Active         |
| 15         | TS09EA1015     | Recycling Truck | 5000.00     | 15        | Active         |
| 16         | TS09EA1016     | Garbage Truck   | 6500.00     | 16        | Inactive       |
| 17         | TS09EA1017     | Mini Tipper     | 1100.00     | 17        | Active         |
| 18         | TS09EA1018     | Compactor Truck | 7500.00     | 18        | Active         |
| 19         | TS09EA1019     | Garbage Truck   | 6000.00     | 19        | Active         |
| 20         | TS09EA1020     | Recycling Truck | 4200.00     | 20        | Maintenance    |
| 21         | TS09EA1021     | Mini Tipper     | 1400.00     | 21        | Active         |
| 22         | TS09EA1022     | Garbage Truck   | 5000.00     | 22        | Active         |
| 23         | TS09EA1023     | Compactor Truck | 8000.00     | 23        | Active         |
| 24         | TS09EA1024     | Recycling Truck | 4800.00     | 24        | Active         |
| 25         | TS09EA1025     | Mini Tipper     | 1000.00     | 25        | Active         |
| 26         | TS09EA1026     | Garbage Truck   | 6500.00     | 26        | Maintenance    |
| 27         | TS09EA1027     | Compactor Truck | 9000.00     | 27        | Active         |
| 28         | TS09EA1028     | Recycling Truck | 4500.00     | 28        | Active         |
| 29         | TS09EA1029     | Garbage Truck   | 5500.00     | 29        | Inactive       |
| 30         | TS09EA1030     | Mini Tipper     | 1200.00     | 30        | Active         |

# Route

```
mysql> SELECT * FROM Route;
```

| Route_ID | Route_Name                   | Area_ID | Vehicle_ID | Collection_Schedule     |
|----------|------------------------------|---------|------------|-------------------------|
| 1        | Miyapur Residential Route    | 1       | 1          | Daily 06:00 AM          |
| 2        | Kukatpally Main Route        | 2       | 2          | Daily 06:30 AM          |
| 3        | Bachupally Residential Route | 3       | 3          | Daily 07:00 AM          |
| 4        | Nizampet Route               | 4       | 4          | Daily 06:30 AM          |
| 5        | Pragathi Nagar Route         | 5       | 5          | Alternate Days 07:00 AM |
| 6        | Chandanagar Route            | 6       | 6          | Daily 06:00 AM          |
| 7        | Madinaguda Route             | 7       | 7          | Daily 07:30 AM          |
| 8        | Hafeezpet Route              | 8       | 8          | Daily 06:30 AM          |
| 9        | Kondapur IT Corridor Route   | 9       | 9          | Daily 05:30 AM          |
| 10       | Madhapur Commercial Route    | 10      | 10         | Daily 05:00 AM          |
| 11       | Gachibowli Route             | 11      | 11         | Daily 05:30 AM          |
| 12       | Manikonda Route              | 12      | 12         | Daily 06:30 AM          |
| 13       | Narsingi Route               | 13      | 13         | Alternate Days 07:00 AM |
| 14       | Khajaguda Route              | 14      | 14         | Daily 07:00 AM          |
| 15       | Jubilee Hills Route          | 15      | 15         | Daily 06:00 AM          |
| 16       | Banjara Hills Route          | 16      | 16         | Daily 05:30 AM          |
| 17       | Ameerpet Route               | 17      | 17         | Daily 06:00 AM          |
| 18       | SR Nagar Route               | 18      | 18         | Daily 06:30 AM          |
| 19       | Moosapet Route               | 19      | 19         | Daily 07:00 AM          |
| 20       | Balanagar Industrial Route   | 20      | 20         | Daily 05:00 AM          |
| 21       | Begumpet Route               | 21      | 21         | Daily 06:00 AM          |
| 22       | Somajiguda Route             | 22      | 22         | Daily 05:30 AM          |
| 23       | Punjagutta Route             | 23      | 23         | Daily 06:00 AM          |
| 24       | Koti Market Route            | 24      | 24         | Daily 04:30 AM          |
| 25       | Abids Commercial Route       | 25      | 25         | Daily 05:00 AM          |
| 26       | Himayatnagar Route           | 26      | 26         | Daily 06:00 AM          |
| 27       | Dilsukhnagar Route           | 27      | 27         | Daily 06:30 AM          |
| 28       | LB Nagar Route               | 28      | 28         | Daily 07:00 AM          |
| 29       | Uppal Route                  | 29      | 29         | Daily 06:30 AM          |
| 30       | Tarnaka Route                | 30      | 30         | Daily 07:00 AM          |

ER Diagram

DB Design

Initial Relational Schema

Filled-in tables in RDBMS

## Waste\_Collection

```
mysql> SELECT * FROM Waste_Collection;
```

| Collection_ID | Resident_ID | Vehicle_ID | Collection_Date | Waste_Type_ID | Quantity_Kg | Collection_Status |
|---------------|-------------|------------|-----------------|---------------|-------------|-------------------|
| 1             | 1           | 1          | 2026-09-01      | 1             | 4.50        | Collected         |
| 2             | 2           | 2          | 2026-09-01      | 3             | 3.20        | Collected         |
| 3             | 3           | 3          | 2026-09-02      | 5             | 2.80        | Collected         |
| 4             | 4           | 4          | 2026-09-02      | 2             | 5.40        | Collected         |
| 5             | 5           | 5          | 2026-09-03      | 6             | 3.75        | Delayed           |
| 6             | 6           | 6          | 2026-09-03      | 7             | 6.20        | Collected         |
| 7             | 7           | 7          | 2026-09-04      | 8             | 2.50        | Collected         |
| 8             | 8           | 8          | 2026-09-04      | 10            | 4.10        | Collected         |
| 9             | 9           | 9          | 2026-09-05      | 11            | 1.80        | Collected         |
| 10            | 10          | 10         | 2026-09-05      | 12            | 0.90        | Collected         |
| 11            | 11          | 11         | 2026-09-06      | 13            | 1.20        | Missed            |
| 12            | 12          | 12         | 2026-09-06      | 14            | 2.30        | Collected         |
| 13            | 13          | 13         | 2026-09-07      | 15            | 1.50        | Collected         |
| 14            | 14          | 14         | 2026-09-07      | 16            | 2.80        | Delayed           |
| 15            | 15          | 15         | 2026-09-08      | 17            | 1.10        | Collected         |
| 16            | 16          | 16         | 2026-09-08      | 18            | 3.60        | Collected         |
| 17            | 17          | 17         | 2026-09-09      | 19            | 2.40        | Collected         |
| 18            | 18          | 18         | 2026-09-09      | 20            | 8.50        | Collected         |
| 19            | 19          | 19         | 2026-09-10      | 21            | 3.90        | Collected         |
| 20            | 20          | 20         | 2026-09-10      | 22            | 2.00        | Missed            |
| 21            | 21          | 21         | 2026-09-11      | 23            | 6.50        | Collected         |
| 22            | 22          | 22         | 2026-09-11      | 24            | 4.30        | Collected         |
| 23            | 23          | 23         | 2026-09-12      | 25            | 5.70        | Delayed           |
| 24            | 24          | 24         | 2026-09-12      | 26            | 7.20        | Collected         |
| 25            | 25          | 25         | 2026-09-13      | 27            | 2.60        | Collected         |
| 26            | 26          | 26         | 2026-09-13      | 28            | 1.90        | Collected         |
| 27            | 27          | 27         | 2026-09-14      | 29            | 3.40        | Collected         |
| 28            | 28          | 28         | 2026-09-14      | 30            | 5.10        | Collected         |
| 29            | 29          | 29         | 2026-09-15      | 4             | 4.80        | Delayed           |
| 30            | 30          | 30         | 2026-09-15      | 9             | 6.70        | Collected         |

ER Diagram

DB Design

Initial Relational Schema

Filled-in tables in RDBMS

## Recycling\_Center

```
mysql> SELECT * FROM Recycling_Center;
```

| Center_ID | Center_Name                        | Location                  | Capacity_Kg | Contact_Number |
|-----------|------------------------------------|---------------------------|-------------|----------------|
| 1         | Miyapur Recycling Center           | Miyapur, Hyderabad        | 10000.00    | 9100000001     |
| 2         | Kukatpally Waste Processing Center | Kukatpally, Hyderabad     | 15000.00    | 9100000002     |
| 3         | Bachupally Recycling Facility      | Bachupally, Hyderabad     | 12000.00    | 9100000003     |
| 4         | Nizampet Recycling Center          | Nizampet, Hyderabad       | 9000.00     | 9100000004     |
| 5         | Pragathi Nagar Waste Center        | Pragathi Nagar, Hyderabad | 11000.00    | 9100000005     |
| 6         | Chandanagar Recycling Facility     | Chandanagar, Hyderabad    | 14000.00    | 9100000006     |
| 7         | Madinaguda Waste Center            | Madinaguda, Hyderabad     | 8000.00     | 9100000007     |
| 8         | Hafeezpet Recycling Center         | Hafeezpet, Hyderabad      | 10000.00    | 9100000008     |
| 9         | Kondapur Recycling Facility        | Kondapur, Hyderabad       | 18000.00    | 9100000009     |
| 10        | Madhapur Waste Processing Center   | Madhapur, Hyderabad       | 20000.00    | 9100000010     |
| 11        | Gachibowli Recycling Center        | Gachibowli, Hyderabad     | 22000.00    | 9100000011     |
| 12        | Manikonda Waste Center             | Manikonda, Hyderabad      | 13000.00    | 9100000012     |
| 13        | Narsingi Recycling Facility        | Narsingi, Hyderabad       | 12000.00    | 9100000013     |
| 14        | Khajaguda Waste Center             | Khajaguda, Hyderabad      | 9000.00     | 9100000014     |
| 15        | Jubilee Hills Recycling Center     | Jubilee Hills, Hyderabad  | 16000.00    | 9100000015     |
| 16        | Banjara Hills Waste Center         | Banjara Hills, Hyderabad  | 17000.00    | 9100000016     |
| 17        | Ameerpet Recycling Facility        | Ameerpet, Hyderabad       | 14000.00    | 9100000017     |
| 18        | SR Nagar Waste Center              | SR Nagar, Hyderabad       | 10000.00    | 9100000018     |
| 19        | Moosapet Recycling Center          | Moosapet, Hyderabad       | 15000.00    | 9100000019     |
| 20        | Balanagar Waste Processing Center  | Balanagar, Hyderabad      | 25000.00    | 9100000020     |
| 21        | Begumpet Recycling Facility        | Begumpet, Hyderabad       | 13000.00    | 9100000021     |
| 22        | Somajiguda Waste Center            | Somajiguda, Hyderabad     | 11000.00    | 9100000022     |
| 23        | Punjagutta Recycling Center        | Punjagutta, Hyderabad     | 16000.00    | 9100000023     |
| 24        | Koti Waste Processing Center       | Koti, Hyderabad           | 19000.00    | 9100000024     |
| 25        | Abids Recycling Facility           | Abids, Hyderabad          | 18000.00    | 9100000025     |
| 26        | Himayatnagar Waste Center          | Himayatnagar, Hyderabad   | 12000.00    | 9100000026     |
| 27        | Dilsukhnagar Recycling Center      | Dilsukhnagar, Hyderabad   | 20000.00    | 9100000027     |
| 28        | LB Nagar Waste Processing Center   | LB Nagar, Hyderabad       | 22000.00    | 9100000028     |
| 29        | Uppal Recycling Facility           | Uppal, Hyderabad          | 17000.00    | 9100000029     |
| 30        | Tarnaka Waste Center               | Tarnaka, Hyderabad        | 14000.00    | 9100000030     |

Waste\_Processing

```
mysql> SELECT * FROM Waste_Processing;
```

| Processing_ID | Collection_ID | Center_ID | Processing_Date | Recyclable_Kg | Organic_Kg | Non_Recyclable_Kg |
|---------------|---------------|-----------|-----------------|---------------|------------|-------------------|
| 1             | 1             | 1         | 2026-09-01      | 1.20          | 2.80       | 0.50              |
| 2             | 2             | 2         | 2026-09-01      | 2.40          | 0.30       | 0.50              |
| 3             | 3             | 3         | 2026-09-02      | 2.10          | 0.20       | 0.50              |
| 4             | 4             | 4         | 2026-09-02      | 0.40          | 4.50       | 0.50              |
| 5             | 5             | 5         | 2026-09-03      | 2.80          | 0.40       | 0.55              |
| 6             | 6             | 6         | 2026-09-03      | 5.00          | 0.70       | 0.50              |
| 7             | 7             | 7         | 2026-09-04      | 2.00          | 0.20       | 0.30              |
| 8             | 8             | 8         | 2026-09-04      | 3.10          | 0.50       | 0.50              |
| 9             | 9             | 9         | 2026-09-05      | 1.20          | 0.30       | 0.30              |
| 10            | 10            | 10        | 2026-09-05      | 0.10          | 0.60       | 0.20              |
| 11            | 11            | 11        | 2026-09-06      | 0.50          | 0.30       | 0.40              |
| 12            | 12            | 12        | 2026-09-06      | 1.70          | 0.30       | 0.30              |
| 13            | 13            | 13        | 2026-09-07      | 0.80          | 0.40       | 0.30              |
| 14            | 14            | 14        | 2026-09-07      | 2.10          | 0.30       | 0.40              |
| 15            | 15            | 15        | 2026-09-08      | 0.20          | 0.70       | 0.20              |
| 16            | 16            | 16        | 2026-09-08      | 1.00          | 2.20       | 0.40              |
| 17            | 17            | 17        | 2026-09-09      | 0.30          | 1.80       | 0.30              |
| 18            | 18            | 18        | 2026-09-09      | 3.20          | 4.00       | 1.30              |
| 19            | 19            | 19        | 2026-09-10      | 2.50          | 0.80       | 0.60              |
| 20            | 20            | 20        | 2026-09-10      | 0.70          | 0.80       | 0.50              |
| 21            | 21            | 21        | 2026-09-11      | 4.00          | 1.60       | 0.90              |
| 22            | 22            | 22        | 2026-09-11      | 2.80          | 1.00       | 0.50              |
| 23            | 23            | 23        | 2026-09-12      | 3.60          | 1.40       | 0.70              |
| 24            | 24            | 24        | 2026-09-12      | 4.80          | 1.70       | 0.70              |
| 25            | 25            | 25        | 2026-09-13      | 1.40          | 0.70       | 0.50              |
| 26            | 26            | 26        | 2026-09-13      | 1.10          | 0.50       | 0.30              |
| 27            | 27            | 27        | 2026-09-14      | 2.20          | 0.70       | 0.50              |
| 28            | 28            | 28        | 2026-09-14      | 2.80          | 1.50       | 0.80              |
| 29            | 29            | 29        | 2026-09-15      | 2.90          | 1.20       | 0.70              |
| 30            | 30            | 30        | 2026-09-15      | 3.80          | 2.00       | 0.90              |

# Complain t

```
mysql> SELECT * FROM Complaint;
```

| Complaint_ID | Resident_ID | Complaint_Date | Complaint_Type     | Description                                                                 | Status              |
|--------------|-------------|----------------|--------------------|-----------------------------------------------------------------------------|---------------------|
| 1            | 1           | 2026-09-01     | Missed Collection  | Waste was not collected from the household on the scheduled date.           | Resolved            |
| 2            | 2           | 2026-09-01     | Overflowing Bin    | Community waste bin near the residence is completely full.                  | Resolved            |
| 3            | 3           | 2026-09-02     | Delayed Collection | Garbage collection vehicle arrived several hours late.                      | Resolved            |
| 4            | 4           | 2026-09-02     | Illegal Dumping    | Waste has been dumped illegally near the residential area.                  | Under Investigation |
| 5            | 5           | 2026-09-03     | Bad Odor           | Strong unpleasant smell is coming from the waste collection point.          | Resolved            |
| 6            | 6           | 2026-09-03     | Missed Collection  | Organic waste was not collected during the scheduled collection round.      | Pending             |
| 7            | 7           | 2026-09-04     | Waste Segregation  | Collection staff did not collect segregated waste separately.               | Resolved            |
| 8            | 8           | 2026-09-04     | Overflowing Bin    | Public garbage container is overflowing and attracting stray animals.       | Resolved            |
| 9            | 9           | 2026-09-05     | Vehicle Issue      | Garbage collection vehicle appears to have broken down during collection.   | Under Investigation |
| 10           | 10          | 2026-09-05     | Missed Collection  | Household waste has remained uncollected for two consecutive days.          | Pending             |
| 11           | 11          | 2026-09-06     | Illegal Dumping    | Construction waste has been dumped on the roadside.                         | Under Investigation |
| 12           | 12          | 2026-09-06     | Delayed Collection | Waste collection was delayed beyond the scheduled time.                     | Resolved            |
| 13           | 13          | 2026-09-07     | Bad Odor           | Waste processing area is producing an excessive foul smell.                 | Resolved            |
| 14           | 14          | 2026-09-07     | Overflowing Bin    | Public dustbin near the market is overflowing.                              | Pending             |
| 15           | 15          | 2026-09-08     | Missed Collection  | Recyclable waste was not collected as scheduled.                            | Resolved            |
| 16           | 16          | 2026-09-08     | Waste Segregation  | Mixed waste was collected despite proper segregation by the resident.       | Under Investigation |
| 17           | 17          | 2026-09-09     | Illegal Dumping    | Plastic waste has been dumped in an unauthorized location.                  | Pending             |
| 18           | 18          | 2026-09-09     | Vehicle Issue      | Collection truck was not available for the scheduled route.                 | Resolved            |
| 19           | 19          | 2026-09-10     | Missed Collection  | Waste collection service was missed in the residential street.              | Resolved            |
| 20           | 20          | 2026-09-10     | Bad Odor           | Garbage accumulated near the apartment entrance is causing bad odor.        | Pending             |
| 21           | 21          | 2026-09-11     | Overflowing Bin    | Waste bin near the commercial area has not been emptied.                    | Resolved            |
| 22           | 22          | 2026-09-11     | Delayed Collection | Collection vehicle arrived late and residents had to wait.                  | Resolved            |
| 23           | 23          | 2026-09-12     | Illegal Dumping    | Electronic waste was dumped near the roadside without authorization.        | Under Investigation |
| 24           | 24          | 2026-09-12     | Waste Segregation  | Organic and recyclable waste were mixed during collection.                  | Pending             |
| 25           | 25          | 2026-09-13     | Missed Collection  | Scheduled waste collection was completely missed.                           | Resolved            |
| 26           | 26          | 2026-09-13     | Vehicle Issue      | Garbage truck experienced a mechanical problem during collection.           | Resolved            |
| 27           | 27          | 2026-09-14     | Overflowing Bin    | Community waste container is overflowing due to increased waste generation. | Pending             |
| 28           | 28          | 2026-09-14     | Illegal Dumping    | Household waste has been dumped in an open plot.                            | Under Investigation |
| 29           | 29          | 2026-09-15     | Delayed Collection | Waste collection vehicle arrived two hours later than scheduled.            | Resolved            |
| 30           | 30          | 2026-09-15     | Missed Collection  | Waste was not collected from the assigned route.                            | Pending             |

30 rows in set (0.007 sec)



**Thank  
You**